NAME: ____ CLASS: _________ INDEX: __________

CATHOLIC JUNIOR COLLEGE
JC2 PRELIM EXAMINATION
Higher 2

Answers

BIOLOGY
9744/01

Paper 1 Multiple Choice

28 AUGUST 2017
1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write and/or shade your name, NRIC / FIN number and HT group on the Answer Sheet in the spaces provided unless this has been done for you.

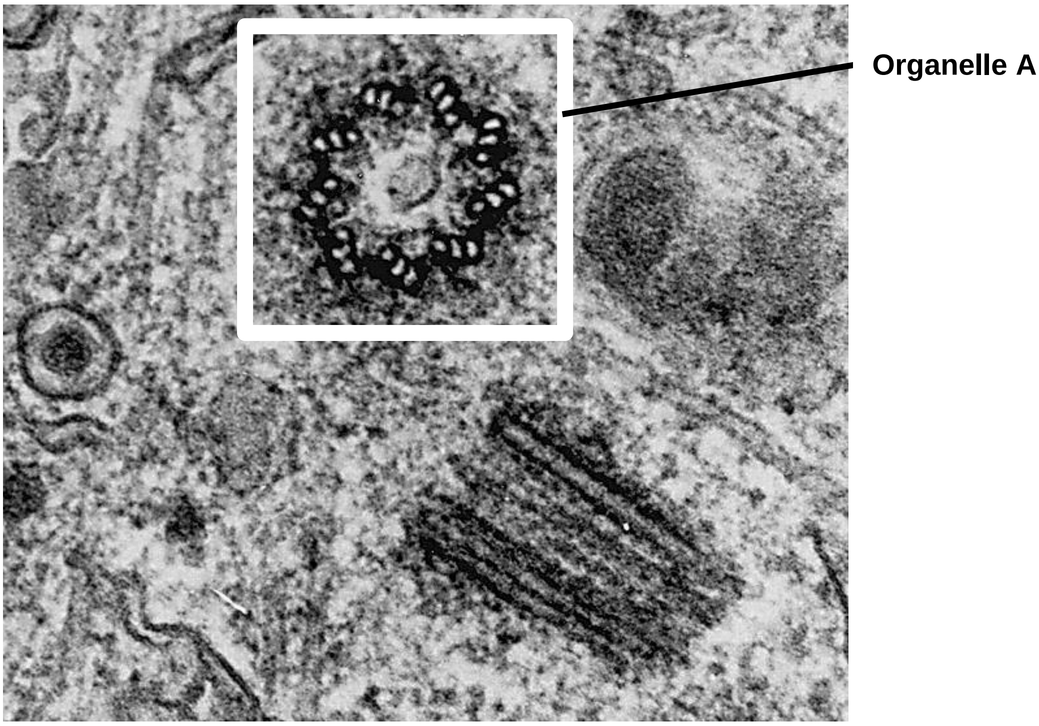
There are **thirty** questions on this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**.
Choose the **one** you consider correct and record your choice in **soft 2B pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
The use of an approved scientific calculator is expected, where appropriate.

1	A	2	C	3	B	4	D	5	B
6	A	7	A	8	C	9	A	10	B
11	C	12	D	13	C	14	A	15	A
16	B	17	A	18	D	19	C	20	C
21	A	22	C	23	C	24	C	25	D
26	B	27	C	28	D	29	B	30	B

1 The figure below shows an electron micrograph of a cross-section of an animal (rat) cell.



Which of the following describes organelle A?

i. 9 triplets of microtubules arranged in a ring.

ii. Inner membrane folded into cristae.

iii. Synthesizes spindle fibres during nuclear division.

iv. Involved in aerobic respiration.

A **i** only

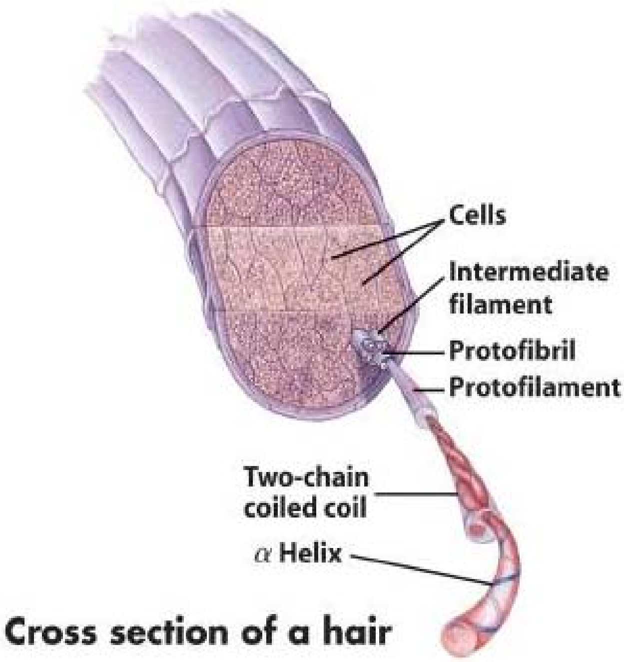
B **i** and **iii** only

C **ii** and **iii** only

D **ii** and **iv** only

ANS A [L1] (H2 SAJC/2016/P1/Q1) [1]

2 Keratin is a fibrous protein in skin, hair and nails. The diagram below shows α-keratin molecules in cross section of a hair follicle.



The features of one form of keratin are listed below:

- **i.** The peptide chain has mainly small amino acid residues.
- **ii.** Each peptide chain forms into an α-helix.
- **iii.** Two helices coil together.
- **iv.** Covalent bonds link adjacent helices.

Which features are different from that of collagen molecules?

A **i** and **ii**

B **i** and **iv**

C **ii** and **iii**

D **iii** and **iv**

ANS C [L1] (H1 A Levels /2014/P1/Q4 modified) [1]

3 How many different types of oligopeptides, each made up of 8 amino acids, may be synthesized using the 20 common amino acids?

A 144 480

B 20^8

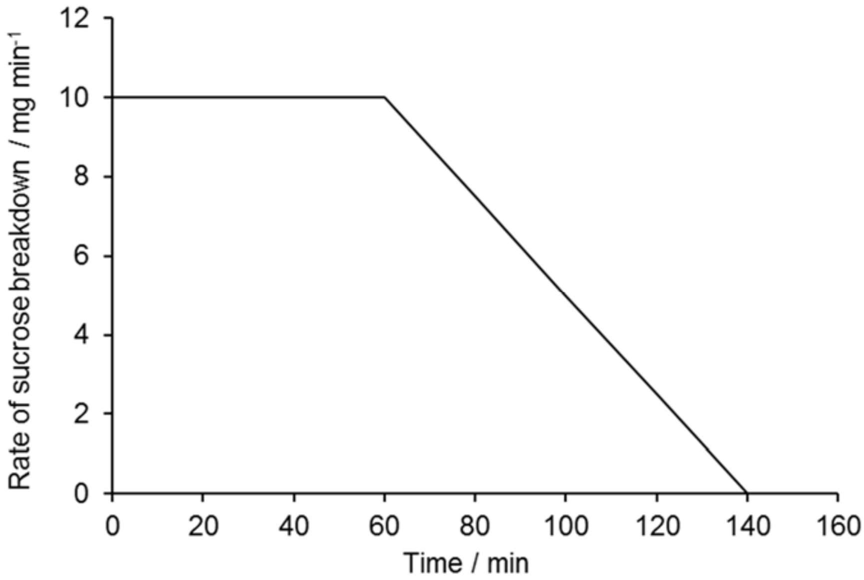
C 8^{20}

D 160

ANS B [L2] (H2 SAJC/2016/P1/Q5 - modified) [1]

4 The graph shows the results of an investigation using invertase, an enzyme that breaks down sucrose into glucose and fructose.

1 g of sucrose was dissolved in 100 cm^3 of water and 2 cm^3 of a 1% invertase solution was added.



Which conclusion can be drawn from this information?

A Between 0 and 60 min, the concentration of the substrate remains constant.

B After 60 min, the concentration of enzymes becomes the limiting factor.

C At 140 min, some of the enzyme molecules are denatured.

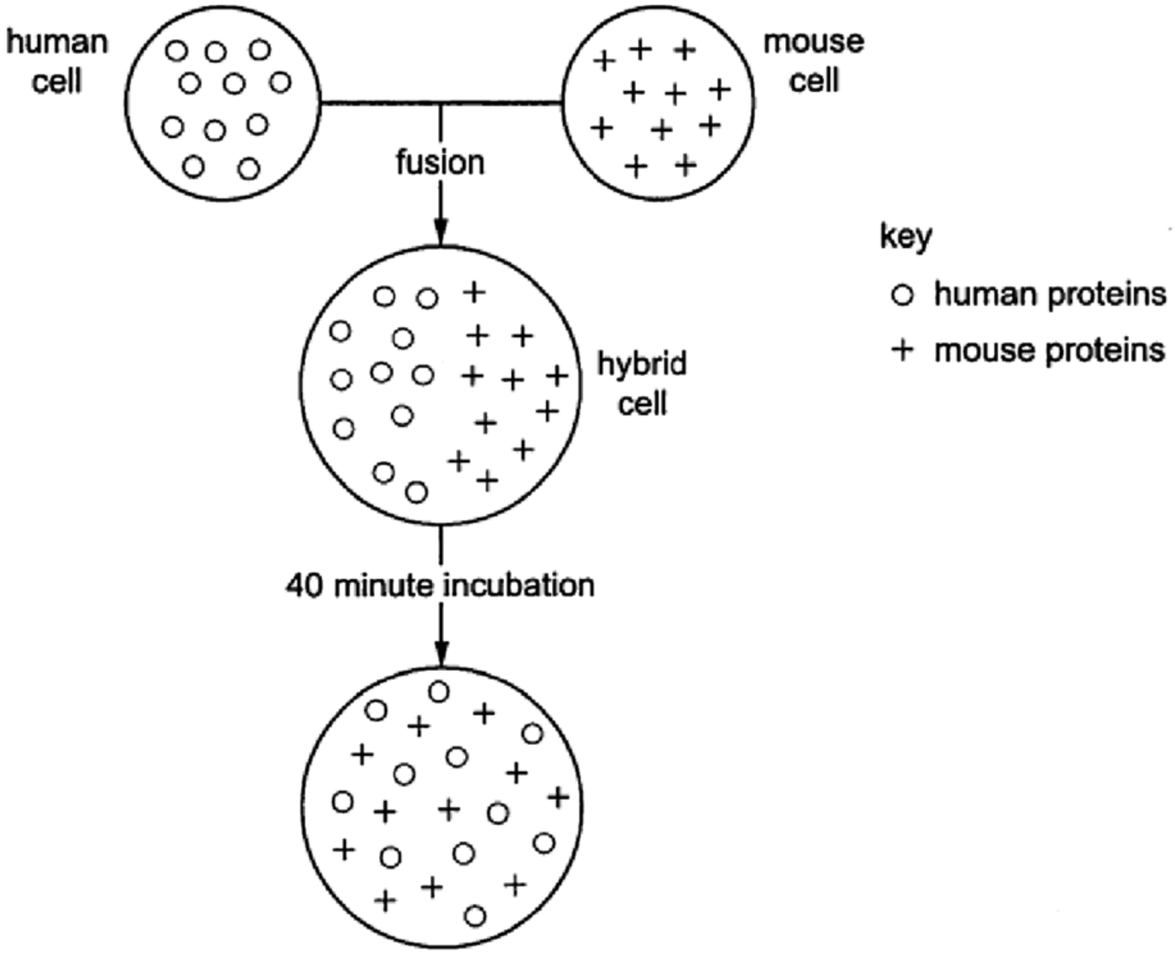
D Between 60 and 140 min, the concentration of the substrate is the limiting factor.

ANS D [L2] (H2 RI/2016/P1/Q6) [1]

5 Human and mouse cells were fused to make hybrid cells. Anti-human and anti-mouse antibodies, carrying different coloured fluorescent dyes, were added. The antibodies bind to the proteins of the cell surface membrane.

The fused cells were incubated for 40 minutes. The locations of the human and mouse membrane proteins were identified at intervals using the fluorescent dyes.

The diagram represents the results of the experiment by showing the positions of the human and mouse proteins on the surface of the cells.



What does this experiment show?

A Movement of the phospholipids pushes the membrane proteins apart.

B Some membrane proteins move through the phospholipids to different places.

C The phospholipids of the human and mouse cell surface membranes do not mix.

D The proteins of human cell surface membranes can move further than those of mouse cells.

ANS B [L2] (H2 A Levels/2015/P1/Q29) [1]

6 Which process involves one stem cell giving rise to two distinct daughter cells: one copy of the original stem cell as well as a second daughter cell programmed to differentiate into a non-stem cell?

A asymmetric replication

B differentiation

C potency

D self renewal

ANS A [L1] (H2 NJC/2016/P1/Q27) [1]

7 DNA replication in eukaryotes involves the following processes.

- RNA primer molecules are attached to each strand at points of origin of replication.
- DNA polymerase attaches to primers and synthesises new strands of DNA in a 5' to 3' direction.
- One strand, called the leading strand, is synthesised in continuous long sections.
- The other strand, called the lagging strand, is synthesised in short sections.
- RNA primers are replaced by DNA nucleotides on both strands.

Which statement explains the difference in the way in which the two strands of a DNA molecule are synthesised?

A DNA polymerase enzymes can only synthesise DNA in one direction.

B Fewer RNA primers are needed on the leading strand.

C The lagging strand has more binding points for RNA primers.

D The replication of DNA is semi-conservative. .

ANS A [L3] (H1 A Levels/2016/P1/Q10) [1]

8 Which statement is **not** true about the transcription in eukaryotes?

A Transcription occurs in the nucleus.

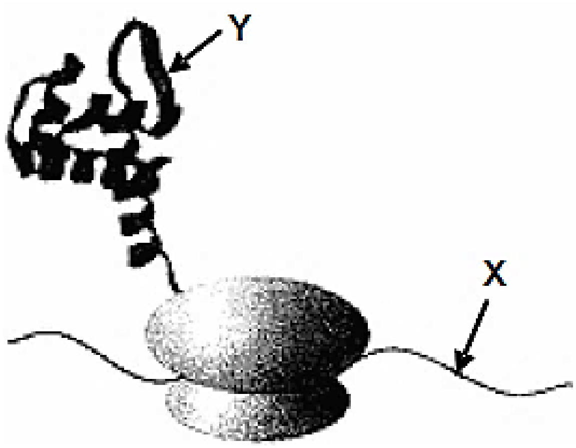
B There are 3 different types of RNA polymerase for the synthesis of mRNA, rRNA, and tRNA.

C The binding of RNA polymerase to TATA box initiates the transcription process.

D No primers are involved during transcription.

ANS C [L1] (Novel) [1]

9 The diagram below shows a particular stage of protein synthesis.



Which of the following statements is true of molecules **X** and **Y**?

A The coiling of molecule Y is a direct result of the information on molecule **X**.

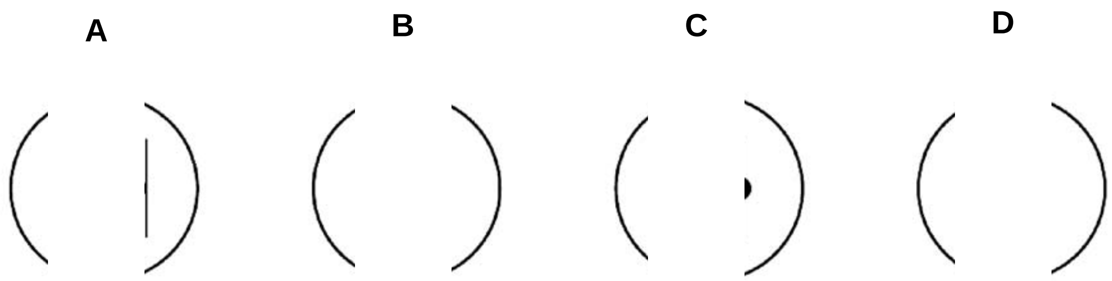
B Molecule **X** is double-stranded whilst molecule **Y** is single-stranded.

C In both molecules **X** and **Y**, the bonds between adjacent monomers of each molecule are phosphodiester bonds.

D The monomers of molecule **X** interact with the monomers of molecule **Y** through temporary hydrogen bonds.

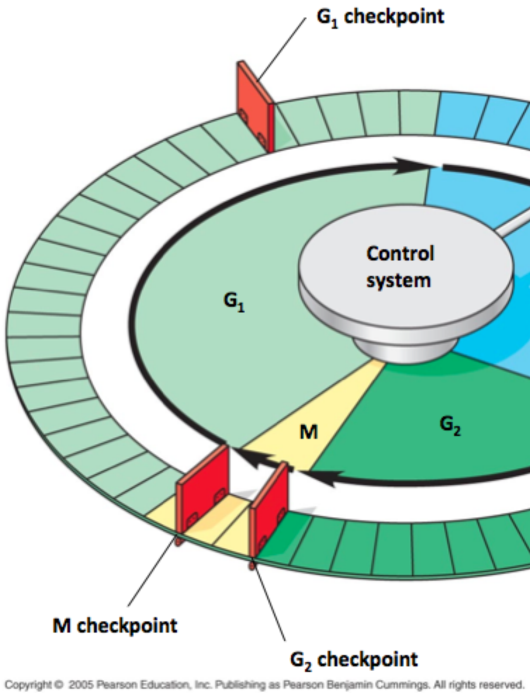
ANS A [L2] (H2 ACJC/2015/P1/Q8) [1]

10 A cell with one pair of chromosomes (2n = 2) undergoes meiosis. Which nucleus is formed at the end of meiosis I?



ANS B [L1] (H2 DHS/2016/P1/Q4) [1]

11 Cell cycle is a highly regulated process. The figure below shows the overview of the different stages and checkpoints in cell cycle.



What happen when G1 checkpoint does not function properly?

A The cell will progress to prophase even when there is mistake during DNA replication.

B The probability of non-disjunction during anaphase increases.

C DNA replication may not occur properly due to the absence of necessary raw materials such as deoxynucleoside triphosphates.

D There will be a decrease in the amount of growth factors secreted.

ANS C [L1] (Novel) [1]

12 What advantages are there in associating eukaryotic DNA with histones to form chromatin?

A Allows large amount of DNA to be packaged into the small space of the nucleus.

B Allows control of gene expression by modulating degree of packaging.

C Protect DNA from degradation which may lead to mutation or death of the cell.

D All of the above.

ANS D [L1] (Novel) [1]

13 Proteins that are ubiquitinated will be transported to proteasome for degradation. Which level of control of gene expression is this?

A Translational level

B Transcriptional level

C Post-translational level

D Post-transcriptional level

ANS C [L1] (Novel) [1]

14 Which of the following statements about bacterial chromosome structure is/are true?

i. Not associated with histone proteins.

ii. Single-stranded chromosome.

iii. Located in the nucleoid region of a nucleus.

iv. Most genes are separated by intergenic DNA sequences.

A **i** only

B **i** and **ii** only

C **ii** and **iii** only

D **ii** and **iv** only

ANS A [L1] (H2 SAJC/2016/P1/Q12) [1]

15 When a mutant strain of Escherichia coli that has lost the regulatory gene of its tryptophan operon is placed in a medium that contains all nutrients the cell need to grow except tryptophan, which of the following will occur?

A The cells will grow even though there is no tryptophan in the medium.

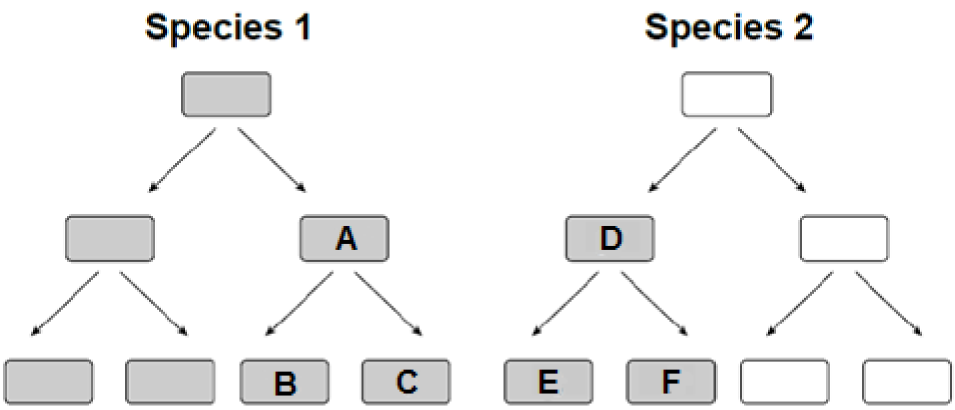
B The cells will grow until excessive tryptophan arrests the expression of the operon.

C The cells will not grow until enough tryptophan has been synthesised to make the repressor active.

D The cells will never grow unless tryptophan is added to the medium.

ANS A [L1] (H2 AJC/2016/P1/Q12) [1]

16 The diagram below shows how two species of bacteria reproduce when placed together in a growth medium. The bacteria that are shaded are resistant to the antibiotic penicillin.



Which one of the following statement(s) is/are likely to be true?

i. Bacteria **B** and **C** are resistant to penicillin as a result of binary fission of Bacterium **A**.

ii. Bacteria **C**, **D** and **F** are resistant to penicillin as a result of random mutation.

iii. Bacterium **D** is resistant to penicillin as a result of conjugation process which transfers the F plasmid carrying penicillin resistance gene from Bacterium **A**.

iv. Bacterium **D** is resistant to penicillin through transduction from Bacterium **A** where there is transfer of the complete F plasmid.

A **iii** only

B **i** and **iii**

C **i** and **iv**

D **ii, iii** and **iv**

ANS B [L2] (H2 AJC/2016/P1/Q11) [1]

17 Which of the following is true of influenza and HIV viruses?

A Genetic shift causes variation in influenza but not in HIV.

B Both HIV and influenza are virulent upon budding off from their respective host cell.

C Influenza viruses have DNA genomes that are templates for transcription.

D HIV viruses have genomes that are readily inserted into the host genome.

ANS A [L2] (Novel) [1]

Please note that Question 18 and 19 are related.

18 Fig 18.1 shows the base sequence of a normal human beta-globin gene and a mutant variant which causes a blood-related disease.

Normal
Non-template strand 5' CAC GTG GAC GGA GGA CTC CTC 3'

Mutant
Non-template strand 5' CAC GTG GAC GGC GGA CAC CTC 3'

Fig. 18.1

Which of the following is correct?

A The blood-related disease is caused by 1 point mutations with more than one amino acid changed.

B The blood-related disease is caused by 1 point mutations with one amino acid changed.

C The blood-related disease is caused by 2 point mutations with more than one amino acid changed.

D The blood-related disease is caused by 2 point mutations with one amino acid changed.

ANS B [L2] (Novel) [1]

19 Fig. 19.1 shows the southern blot of the co-dominant blood-related disease shown in Fig. 18.1. Only one restriction enzyme *MstII* was used and the blot was hybridized with a probe specific for the beta-globin gene.

Individual	**1**	**2**	**3**	**4**	**5**
Phenotype	**lives**	**lives**	**lives**	**dies**	**lives**

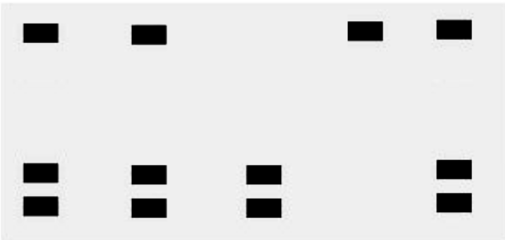


Fig. 19.1

With reference to Fig. 19.1 and Fig. 18.1; which of the following statements is correct?

A Individuals 1, 2 and 5 are normal.

B Individuals 1, 2 and 5 are homozygous at the loci for beta-globin gene.

C Individuals 4 and 5 are homozygous and heterozygous at the loci for beta-globin gene respectively.

D Individuals 3 and 4 are homozygous and heterozygous at the loci for beta-globin gene respectively.

ANS C [L2] (Novel) [1]

20 A tall green stemmed plant with genotype TTrr was crossed with a short red stemmed plant with genotype ttRR. The F1 plants were allowed to self fertilise. A X^2 test was carried out on the results obtained for the F2 generation. Part of the values for X^2 are shown:

Deg. of freedom	p= 0.5	p= 0.1	p= 0.05	p= 0.01	p= 0.001
1	0.46	2.71	3.84	6.64	10.83
2	1.39	4.6	5.99	9.21	13.82
3	2.37	6.25	7.82	11.34	16.27
4	3.36	7.78	9.49	13.28	18.46
5	4.35	9.24	11.07	15.09	20.52

The value of X^2 was 7.6 in this investigation.

What is the probability of this value of X^2 and do the results fit the expected ratio?

	Probability	*Results fit expected ratio*
A	Between 0.01 and 0.05	No
B	Between 0.01 and 0.05	Yes
C	Between 0.05 and 0.1	Yes
D	Between 0.05 and 0.1	no

ANS C [L1] (N2005/1/22 modified) [1]

21 Which of the following is true about gene interactions?

A Gene interactions involve two genes controlling one character.

B Gene interactions involve masking and follow mendelian phenotypic ratios.

C Gene interactions are a form of co-dominance

D Gene interactions restrict the number of phenotypes seen.

ANS A [L1] (Novel) [1]

22 Sodium azide is a strong inhibitor to the ETC in mitochondria. An experiment was conducted with isolated mitochondria, 2 molecules of glucose, 10 molecules of pyruvic acid along with oxygen, ADP and NAD^+ which was supplied in excess.
What would be the expected production of ATP from the experiment?

A 2

B 4

C 10

D 22

ANS C [L3] (Novel) [1]

23 Which of the following is not true of photosynthesis.

A The spectra of light in which photosynthesis is most efficient is in the red and violet region.

B Rate of ATP synthesis depends on the differential proton gradient across the thylakoid membrane.

C Photosynthesis starts first with the photolysis of water.

D Oxygen concentration affects the efficiency of the light independent reaction.

ANS C [L2] (Novel) [1]

24 The Fig. 24.1 represents a G-protein coupled receptor on the cell surface membrane. A peptide hormone ligand is bound to the receptor and initiates the production of a second messenger.

Fig. 24.1

What is the second messenger?

A a peptide hormone.

B ATP

C cyclic AMP

D Kinase

ANS C [L1] (2007 9744 P1.Q30) [1]

25 Birds, such as cockatoos, have a species of louse (an insect parasite) that lives on their feathers.

White, sulfur-crested cockatoos have pale lice on their wings and bodies while yellow-tailed black cockatoos have dark lice on their wings and bodies. Both of these cockatoos have black lice of this species on their heads. In order to rid themselves of these parasites, cockatoos preen their wings and bodies with their beaks but have to use their feet to preen their heads.

What best explains how this species of louse has diversified into two colour variants on the birds' wings and bodies, but has remained dark on the birds' heads?

A Cockatoo beak preening results in selection pressure on wing and body lice.

B Cockatoos are unable to see the lice while preening their heads.

C Cockatoos notice badly camouflaged lice on their wings and bodies while preening.

D Cockatoos use different preening techniques on different parts of their bodies resulting in natural selection.

ANS D [L2] (H1 A Levels/2015/P1/Q23) [1]

26 Bacteria in the genus *Wolbachia* infect many butterfly species. They are passed from one generation to the next in eggs, but not in sperm, and they selectively kill developing male embryos.

In Samoa in the 1960s, the proportion of male blue moon butterflies fell to less than 1% of the population. However, by 2006, the proportion of males was almost 50 % of the population.

Resistance to *Wolbachia* is the result of the dominant allele of a suppressor gene.

Which statements correctly describe the evolution of resistance to *Wolbachia* in the blue moon butterfly population?

i. *Wolbachia* acts as a selective agent.

ii. The selective killing of male embryos is an example of artificial selection.

iii. When infected with *Wolbachia,* male embryos that are homozygous for the recessive allele of the suppressor gene die.

iv. All male embryos that carry the dominant allele of the suppressor gene pass that allele to their offspring.

v. The frequency of the dominant allele of the suppressor gene rises in the butterfly population.

A **i** and **iv**

B **i**, **iii** and **v**

C **ii** and **iii**

D **ii**, **iv** and **v**

ANS B [L2] (H1 A Levels/2011/P1/Q23) [1]

27 Which of the following statements correctly relate to molecular phylogenetics?

i. Lines of descent from a common ancestor to present-day organisms have undergone similar, fixed rates of DNA mutation.

ii. Organisms with similar base sequences in their DNA are closely related to each other.

iii. The number of differences in the base sequences of DNA of different organisms can be used to construct evolutionary trees.

iv. The proportional rate of fixation of mutations in one gene relative to the rate of fixation of mutations in other genes stays the same in any given line of descent.

A **i** and **ii**

B **i** and **iv**

C **ii** and **iii**

D **iii** and **iv**

ANS C [L2] (H2 VJC/2015/P1/Q31) [1]

28 Which statement about vaccination is true?

A Vaccination of a small proportion of the population can break the disease transmission cycle.

B Vaccination can prevent and control disease, but it is unable to eradicate the disease.

C Vaccination stimulates body's innate immune system, thus protecting the individual from future infection by the same pathogen.

D Vaccination stimulates immunity without causing the disease.

ANS D [L1] (Novel) [1]

29 Fig 29.1 illustrating the effects of elevated CO_2 on growth and development of soybean.

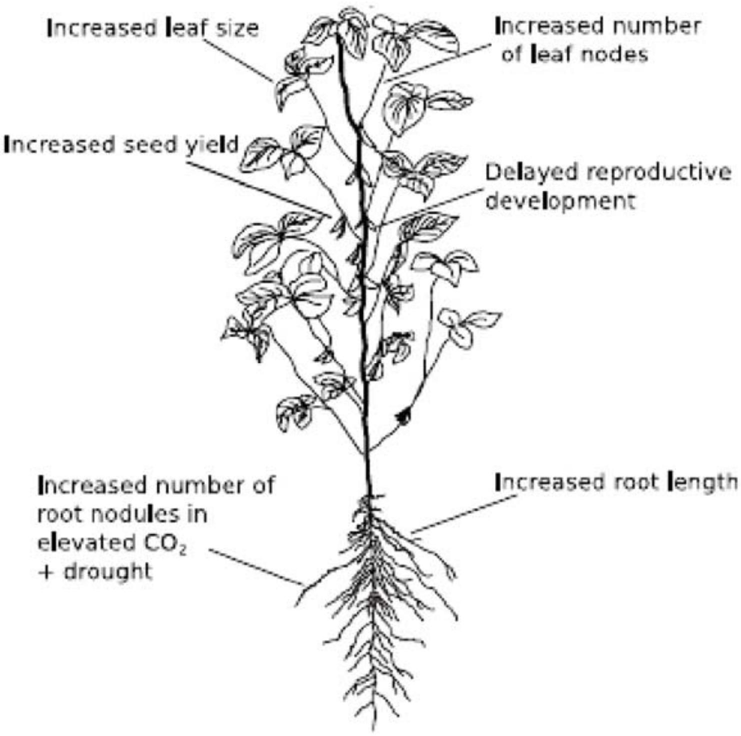


Fig. 29.1

With reference to Fig. 29.1. Which of the following can be inferred from the elevated levels of CO_2.

A Plant photosynthetic rates will increase as CO_2 levels increase.

B Plant biomass increases but dispersal range decreases.

C Plant respiration will outweigh that of photosynthesis.

D Plants are now better able to ensure their own survival and continuation.

ANS B [L2] from L1 (Novel) [1]

30 Which of the following is not true about climate change and biodiversity?

A Climate change results in specific selection pressures that may be disadvantageous to most species causing a decrease in biodiversity.

B Climate change over a short time may result in different selection pressures which may promote speciation and promote biodiversity.

C Climate change may negatively affect keystone species which then affect ecosystems eventually affecting biodiversity.

D Climate change may cause a lowering of temperatures and may push species to physiological limits and eventually lower biodiversity.

ANS B [L2] (Novel) [1]

END OF PAPER